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Current and projected impacts of extreme climate events on winter wheat yield in Northern China

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Extreme climate events (ECEs) pose substantial threats to global food security, particularly impacting winter wheat yields in Northern China. Here, we examine historical and projected effects of ECEs on wheat production across three sub-regions. Using statistical models and cropping system simulations, we quantify yield losses driven by individual and compound stressors under historical and future climate scenarios. Historically, post-anthesis compound hot-dry events and spring heat stress caused the greatest yield reductions in the north and middle sub-regions, whereas compound freeze-dry events dominated in the south. By 2080-2100, rising post-anthesis compound hot-dry events are projected to further reduce yields. Heat stress is expected to offset the benefits of reduced freeze stress, particularly in the cooler north sub-region, where its effects are more pronounced than in the warmer south. These findings highlight the urgent need for region-specific adaptation strategies to sustain wheat production under climate change.

As the global population and living standards drive the increasing demand for food, projections suggest that food production will need to double by 2050[1,2]. However, climate change poses challenges to food security with changes in temperature and precipitation patterns, substantially affecting crop growth environments and final yields[3]. Under continued greenhouse gas emissions, the frequency and intensity of extreme climate events (ECEs) are expected to increase, exerting predominantly negative impacts on agricultural systems and likely causing yield reductions for many major crops[4,5]. Consequently, understanding the interactions between key climate factors and their impacts on crop productivity is critical for developing strategies to mitigate climate-induced risks to agriculture.

ECEs cause large variations in daily and seasonal temperature and precipitation, and these occurrences often cause varying degrees of yield losses. Wheat is most vulnerable to heat stress during the reproductive phase from anthesis to maturity[6,7]. Heat stress affects crop growth and yield by reducing photosynthetic rate and carbon assimilation[8], shortening grain filling duration[9], and reducing grain number[10]. Freeze stress likely affects crop growth and development, thereby reducing production[11,12]. A sudden cold temperature exposure during the sensitive stem elongation phase can reduce the number of tillers and spikes. As shown in a two-year field experiment, wheat yield loss from spring freeze stress is highly correlated with the reduction of tiller and spike number[13]. A five-day spring freeze episode at the jointing stage could lead to a 5–14% wheat yield loss. In addition, drought is often the main constraint on winter wheat yields in rainfed systems[4]. Drought can restrain root growth and leaf expansion, shorten grain filling duration, and limit carbon allocation in crops[14,15]. Long-term drought can cause complete crop failure and can sharply reduce global crop yields by an average of 10%[4]. What makes this more challenging is that various ECEs can interact during the critical growing stages, namely, compound ECEs can lead to unique and even greater negative impacts than individual events[16]. Such interactions can increase the susceptibility of one ECE to subsequent deleterious environmental conditions, ultimately intensifying final crop yield reduction.

Two distinct methods have been widely used to examine relationships between climate and yield: statistical models and process-based crop models. Statistical models use various regression methods to quantify the impacts of changes in climate indices on crop yield[17–20]. However, a major limitation of statistical models is that they usually provide a simple evaluation of impacts, rather than revealing the deeper underlying mechanisms required to better inform adaptation strategies[18,21]. Process-based crop models are essential tools designed to assess the effects of multiple climate factors on crop growth and yield[22–24]. They can dynamically simulate the process of crop growth, development, and yield formation, and accurately express the complex interactions between crop growth, environmental factors, and management practices. However, most crop models have been developed under relatively stable climate conditions. The effects of ECEs on

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crops are generally oversimplified and only vaguely described, leading to inaccurate simulation results under extreme climate conditions[18,25]. Given the limitations of both process-based crop models and statistical regression models, the value of combining both models to achieve complementary advantages is gaining recognition as a more robust approach to determining the impacts of ECEs on crop performance and production.

Winter wheat is a major crop in the Huang-Huai-Hai (HHH) Plain of China (Fig. 1a), contributing 21% of global food supplies, and 80% of China's wheat production[26], making it important for both domestic and international food security. The HHH wheatbelt is characterized by variable climate conditions, featuring north-south gradients in both temperature and precipitation across the region. Average growing season temperatures in HHH had a warming trend occurring across the entire region over the past 40 years (Fig. 1b) but presented a declining trend in average total precipitation (Fig. 1c). ECEs, including heat, freeze, and drought, frequently impact China's agricultural production. Xiao et al. (2018)[27] reported an increasing probability of spring freeze since the 1970s, while Liu et al. (2014)[28] projected that drought and heat stress will intensify under a changing climate in Northern China. These recurrent ECEs are expected to amplify the interannual variability of wheat yields, posing a severe threat to food security. While numerous studies have examined the relationship between ECEs and crop production in China[29,30], a systematic analysis and modeling of how different ECEs affect wheat yields across various biological growth phases remains unexplored. There is thus a need to elucidate the linkage between ECEs over the agricultural phases of relevance to crop growth. Moreover, understanding how these impacts may evolve under future climate scenarios is critical. Therefore, to examine the relationship more fully between climate and wheat productivity, the HHH was divided into three wheat planting sub-regions based on the eco-climatic conditions, delineated as the north (NS), middle (the Huang-Huai, HHS), and south (the middle-lower reaches of Yangzi River, MYS). Our study aims to (1) assess associations between historical ECEs and wheat yield losses during different crop growth phases in the three main wheat-growing sub-regions of China, and (2) evaluate the potential impacts of ECEs on future wheat yields under climate scenarios of SSP1-2.6, SSP3-7.0, and SSP5-8.5.

Results

Response of wheat yield to extreme climate indices

We use a linear mixed-effects model that includes freezing (FDD), heat (EDD), dry (CDD), compound hot-dry (CHD), and compound freeze-dry (CFD) events to estimate the sensitivity of wheat yield to different extreme climate indices in the winter wheatbelt (Fig. 2). Due to the limited availability of observed yields, regression models were applied for extreme climate indices spanning from 2000 to 2019. In general, the coefficients for heat stress were larger in magnitude than for freezing stress, with heat stress during the spring (EDD_{VP}) having a stronger negative impact on yield (0.44%) than during the post-anthesis phase (EDD_{RP}) (0.08%) (Fig. 2a). The estimated sensitivity of wheat yield to consecutive dry events (CDD) was negative during the vegetative and reproductive phases, with higher values corresponding to a greater sensitivity of yield to CDD (Fig. 2b). Among all the CDD terms, an increase of dryness during the reproductive (CDD_{RP}) phase was associated with the largest yield loss, with the increase in CDD_{RP} reducing yield up to 4.1%. We found there was a consistent negative response to yield when single extreme climate indices were included in the compound extreme climate indices (Fig. 2c). We observed a decreased resilience of yields to extreme climate indices over time, with a trend of −0.06 kg % yr^{-1} from 2000 to 2009. Notably, the period from 2010 to 2019 experienced a relatively higher yield decline of −0.17 % yr^{-1}, which had almost double the negative effect on yield compared to that from 2000 to 2009.

To understand what drives yield changes for each sub-region, we decomposed yield into losses due to freezing, heat, dry, compound hot-dry, and compound freeze-dry events. The results showed that the variation in yields could be explained by climate indices, with distinct negative effects observed across the three sub-regions (Fig. 3, Supplementary Table 1). In the north (NS), heat stress during the reproductive phase (EDD_{RP}) and dryness during the vegetative phase (CDD_{VP}) caused comparable wheat yield loss. Meanwhile, the heat stress during the spring (EDD_{VP}) led to a larger yield loss, with the increase in EDD_{VP} reducing yield by 6.4% (Fig. 3a). This was consistent with previous findings that high temperatures in the spring just before or during flowering affect seed set, cause accelerated leaf senescence, and result in larger declines in wheat yield[31]. Also, the yield loss caused by dry events during the vegetative phase (CDD_{VP}) was greater than that

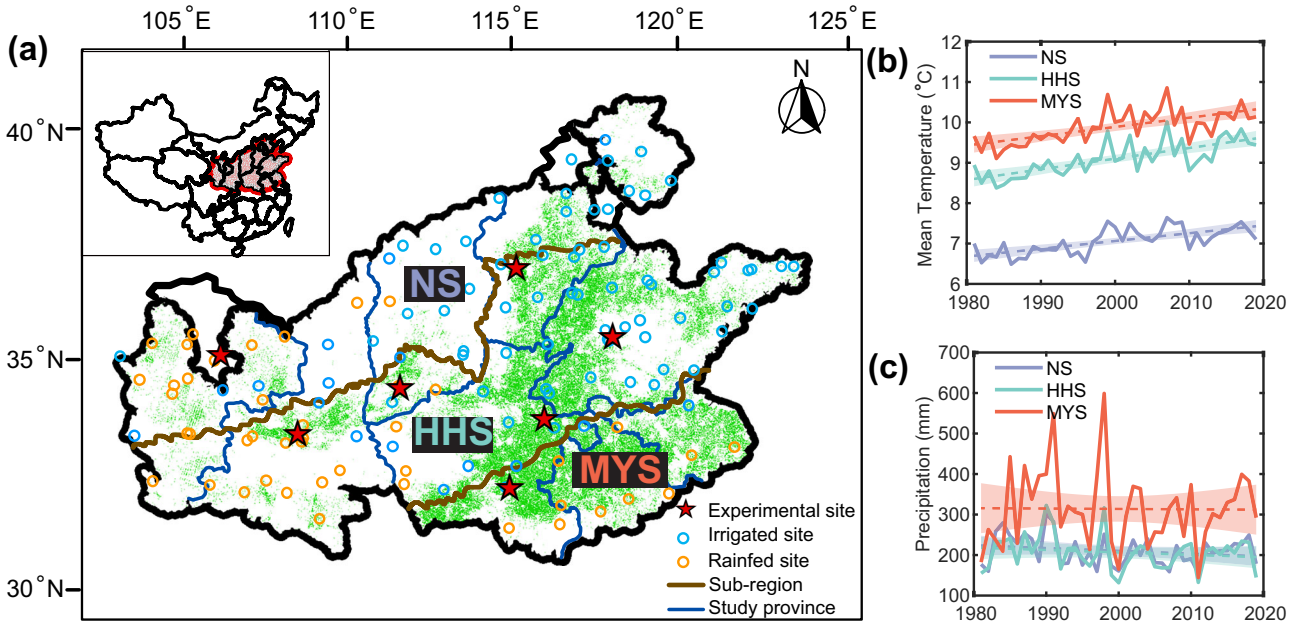


Fig. 1 | Characterization of the study region. a Winter wheat planting patterns (green background) across each sub-region (north (NS); middle (Huang-Huai, HHS); south (middle-lower reaches of Yangzi River, MYS)). Circles are meteorological observations distributed at irrigated (blue) and rainfed (orange) sites, and red stars indicate locations of experimental sites. **b, c** Time series of mean (solid lines) of the winter wheat growing season temperature (**b**) and precipitation (**c**) during 1980 to 2019. The dashed lines in (**b**) and (**c**) represent the linear trends, with shaded areas representing 95% confidence intervals.

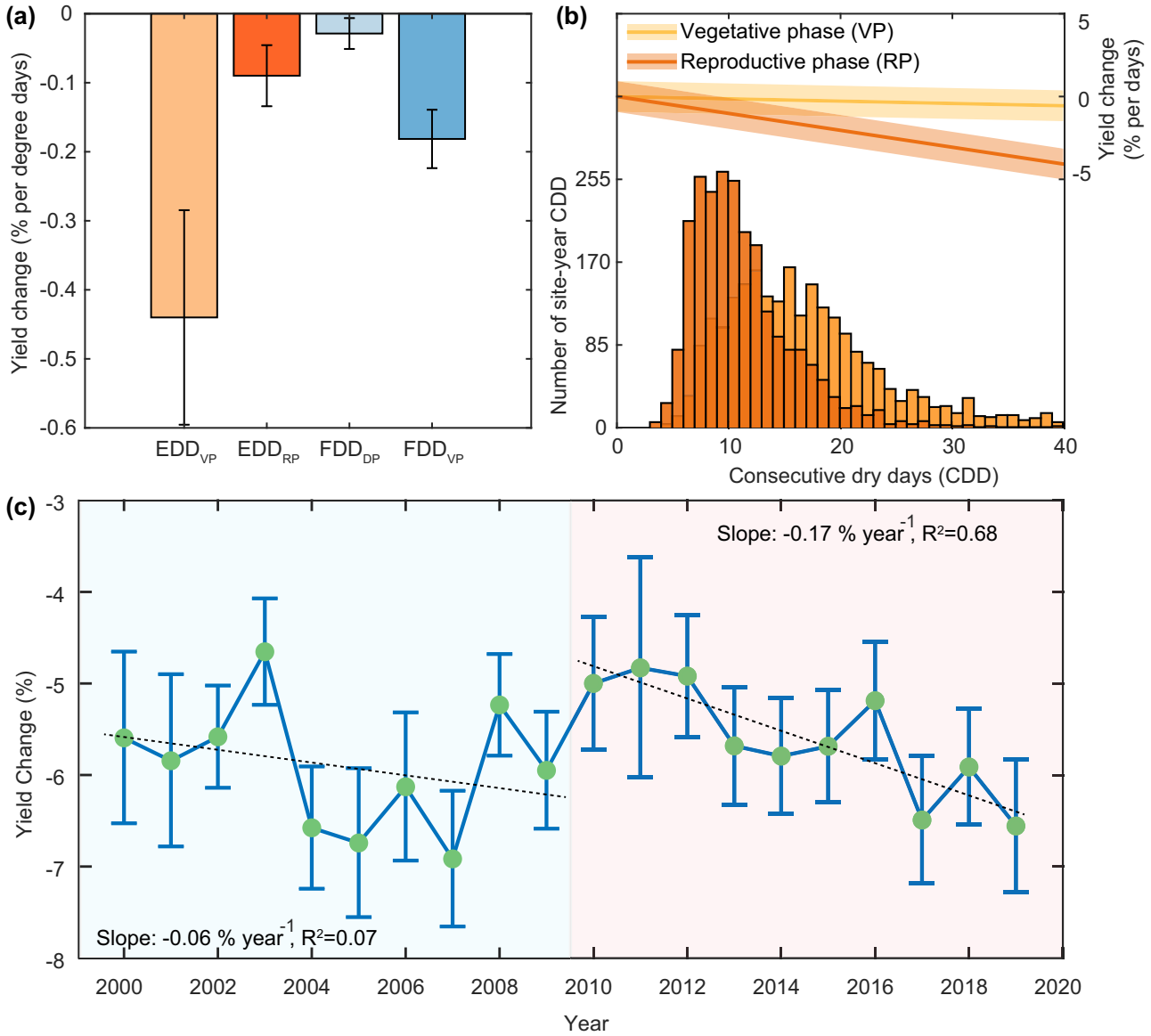


Fig. 2 | Impacts of extreme climate indices on wheat yields. a Sensitivity of winter wheat yields to extreme degree days (EDD) and freezing degree days (FDD) in different phenological phases; the error bars represent 95% confidence intervals. DP is the dormancy phase (December to February); VP is the vegetative phase (March to Anthesis); RP is the reproductive phase (Anthesis to Maturity). **b** Percent change in yield as a function of the consecutive dry days (CDD) by different growth phases; shaded areas indicate the 95% confidence intervals and histograms show the distribution of CDD occurrence amounts sampled. **c** Changes in sensitivity of wheat yields to all extreme climate indices between 2000 to 2009 (blue shaded) and 2010 to 2019 (pink shaded). The solid green circles are estimated mean coefficients from the statistical model, with bars indicating the 95% confidence intervals.

caused by dry events during the other two phases. Compound hot-dry events (CHD_{RP}) caused a considerably larger yield loss of 7.5% compared to individual counterparts. In the middle (HHS), there was no difference in the yield loss caused by compound hot-dry extremes (CHD_{RP}) and heat stress (EDD_{VP}) (Fig. 3b). The average yield loss caused by consecutive dry days (CDD) in the HHS was lower than that in the NS sub-region. However, compound freeze-dry events (CFD) tended to induce greater stress on wheat yield in the south sub-region (MYS), resulting in a yield loss of 12.0% during the dormancy (CFD_{DP}) and 2.5% during the vegetative (CFD_{VP}) phases (Fig. 3c). A weakening sensitivity of yield to compound hot-dry events (CHD_{RP}) was observed in the south compared to the other two sub-regions, and the negative effect of CHD_{RP} found by our results was not present.

Growing season baseline and future climate conditions

Substantial variations of extreme climate indices during wheat growth phases were observed across the winter wheatbelt (Fig. 4). The time series of ECEs was calculated (Supplementary Figs. 1 and 2) and found that future heat events were projected to largely increase. By the 2080s, spring heat events (HD_{VP}) were projected to increase under the three climate scenarios from 1 to 2 days (SSP1–2.6), 4 days (SSP3–7.0), and 6 days (SSP5–8.5). Heat events during the reproductive (HD_{RP}) were expected to increase by an average of 5 days (SSP1–2.6), 11 days (SSP3–7.0), and 16 days (SSP5–8.5), indicating heat especially during the reproductive period, as warming increased (Supplementary Fig. 3). The strongest increases in HD_{VP} were projected to occur across the north and middle sub-regions (Fig. 4b–d). Increases in HD_{RP} showed some spatial variability but were stronger in all sub-regions under most climate scenarios (Fig. 4g, h). During the baseline period, the spatial variations in freeze events (FD) were similar, with higher freeze events in the north (NS) compared to the middle (HHS) and south (MYS) sub-regions. By the 2080s, Freeze events were projected to decrease with the largest of 19 days during the dormancy (FD_{DP}) and 10 days during the vegetative (FD_{VP}) phases under the SSP5-8.5 scenario (Supplementary

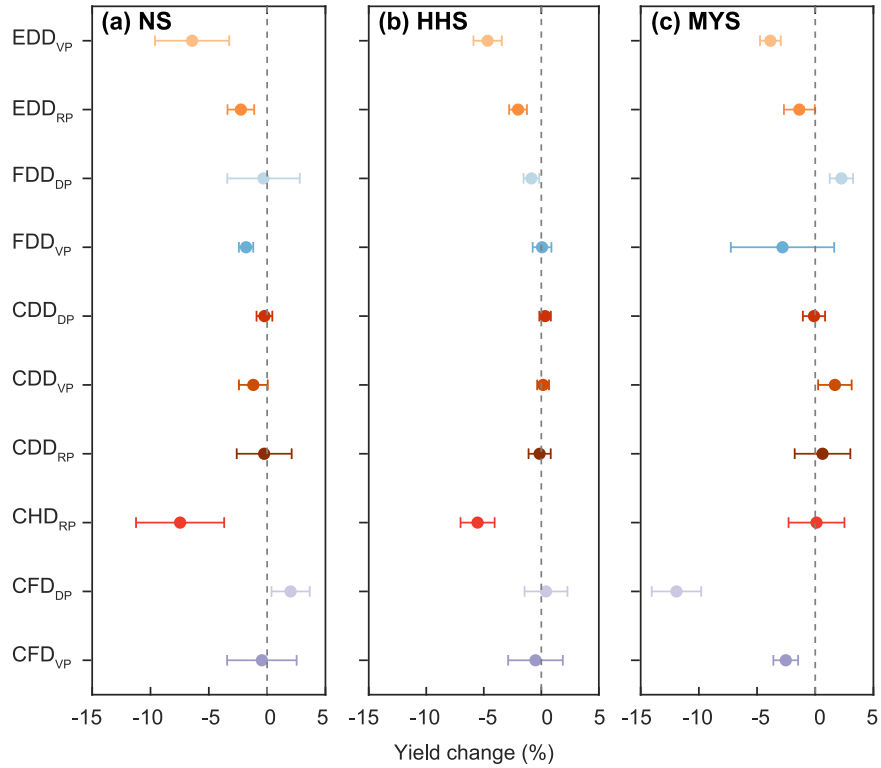


Fig. 3 | Impacts of extreme climate indices on wheat yields at three sub-regions. a Yield changes due to extreme climate indices in the north (NS) sub-region, including extreme degree days (EDD), freezing degree days (FDD), consecutive dry days (CDD), compound hot-dry days (CHD), and compound freeze-dry days (CFD). DP is the dormancy phase (December to February); VP is the vegetative phase (March to Anthesis); RP is the reproductive phase (Anthesis to Maturity). **b, c** Same as (**a**) but for (**b**) middle (Huang-Huai, HHS), and (**c**) south (middle-lower reaches of Yangzi River, MYS) sub-regions. The solid-colored circles represent the estimated mean coefficients for each extreme climate index from the statistical model, and horizontal bars indicate the 95% confidence intervals.

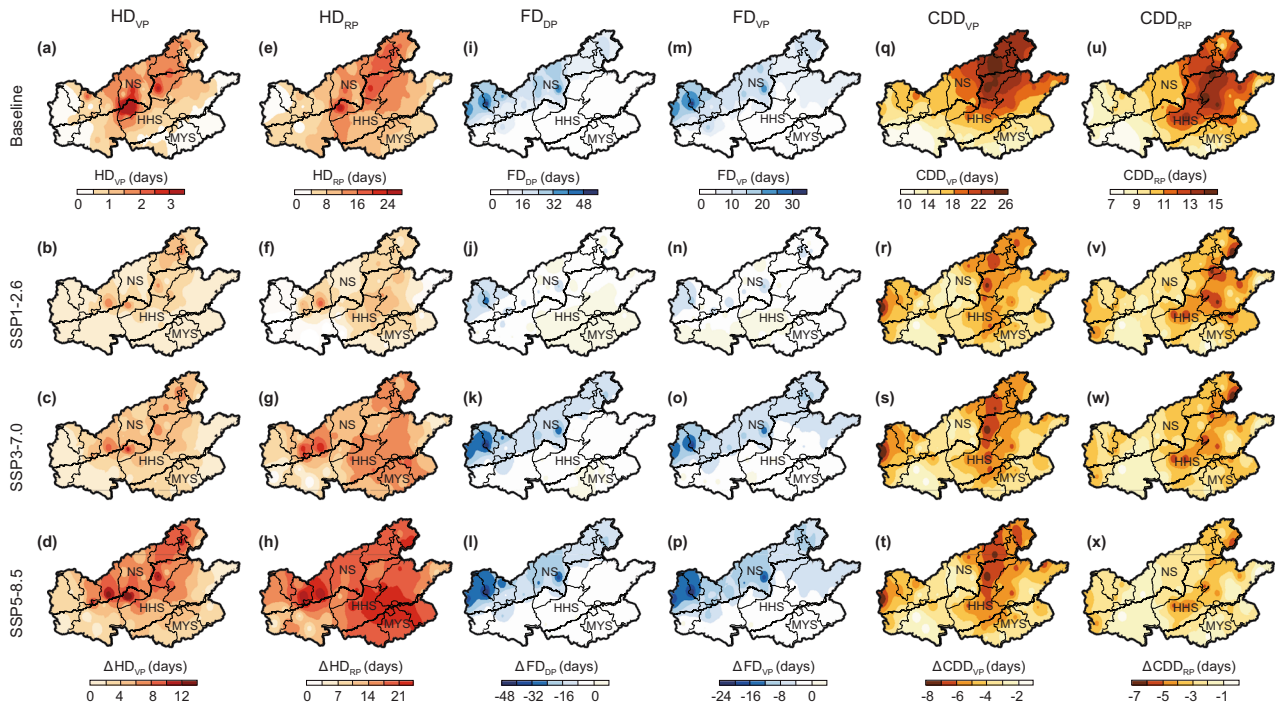


Fig. 4 | Historical and projected changes in single extreme climate indices. a–h Baseline (1999-2019) and projected changes (Δ, 2080-2100) in hot days (HD) during growth phases for each climate scenario. **i–p** Same as (**a–h**) but for freezing days (FD). **q–x** Same as (**a–h**) but for consecutive dry days (CDD). Future climate projections are based on the ensemble of climate models. DP is the dormancy phase (December to February); VP is the vegetative phase (March to Anthesis); RP is the reproductive phase (Anthesis to Maturity).

Fig. 3). Regionally, FD_{DP} was projected to decrease more than FD_{VP} for three climate scenarios across nearly all of NS. Nonetheless, future FD was projected to increase slightly over the central areas of HHS and MYS under the SSP1-2.6 scenario (Fig. 4j and n). These changes indicated that climate change poses more substantial effects on reducing freeze events in the north sub-regions than in the south sub-regions.

The strongest dry events (CDD) were mainly concentrated in the northeastern area of NS and HHS during the baseline period (Fig. 4q, u). Future dry events during the vegetative phase (CDD_{VP}) showed a widespread decline, although spatial variations were apparent. Future dry events during the reproductive phase (CDD_{RP}) showed a smaller average decrease of 3, 3, and 2 days by the 2080s under SSP1-2.6, SSP3-7.0, and SSP5-8.5 pathways, respectively (Supplementary Fig. 3). Larger projected decreases in CDD_{RP} were observed in SSP1–2.6 and SSP3-7.0 compared to SSP5-8.5, which contrasted with the trends of HD and FD, indicating the sensitivity of different ECEs to various emission scenarios. Additionally, the spatial patterns of future CDD differed between the VP and RP periods, with the northwestern and central NS and HHS showing the greatest decreases in CDD_{VP}, while the largest projected decreases in CDD_{RP} were located in central HHS (Fig. 4v–x).

Regarding the compound ECEs, compound hot-dry events (CHD_{RP}) were more prevalent in the north (NS and HHS) sub-region than in the south (MYS) sub-region during the baseline period (Fig. 5). CHD_{RP} was projected to show continuing increases of 9.6% (SSP1–2.6), 31.0% (SSP3–7.0), and 58.2% (SSP5–8.5) by the 2080s (Supplementary Fig. 4). Moreover, future CHD_{RP} exhibited an apparent spatial pattern with greater increased rates found in the central HHS and MYS (Fig. 5b–d). CHD_{RP} was projected to decrease over the western NS and HHS under SSP1-2.6 and SSP3-7.0, but increased profoundly under SSP5–8.5. Compound freeze-dry events (CFD) were projected to show a widespread decline by the 2080s. The spatial variation in CFD was similar to that of FD during the baseline period, although clear spatial differences were still apparent. Future compound freeze-dry events during the dormancy (CFD_{DP}) and vegetative (CFD_{VP}) in NS showed the strongest decreases (Fig. 5f–h). However, the change in future CFD showed increased rates over the central areas of HHS and MYS under the SSP1-2.6 scenario (Fig. 5j–l).

Statistical and process-based modeling of extreme climate indices on wheat yield

Comparison of ECEs estimated from crop and statistical analysis modeling revealed that crop models could reproduce the negative effect of extreme climate indices on yield derived from statistical modeling (Fig. 6). Estimated sensitivity of yield to higher temperatures (EDD) suggested that crop models could better account for heat stress effects on wheat yield (Fig. 6a, b). It should be noted that relatively large discrepancies existed in freezing stress between statistical and process-based modeling (Fig. 6c). Process-based crop models might not adequately simulate the growth and development dynamics, and yield formation of wheat under low-temperature stress, suggesting possible flaws or gaps in model representation of the freezing stress effects on crop yield[10,32]. This may arise because crop phenology sub-models were developed with field experiment data, which were scarce under extreme conditions, especially extreme low temperature conditions. For yield changes due to drought effects, crop models generally reproduced the direction of yield change, but the magnitude of yield change estimated by crop models differed substantially from the one estimated from yield statistics (Fig. 6f). Interestingly, estimated sensitivity of yield to compound hot-dry events (CHD_{RP}) showed a larger difference between observed and simulated models than compound freeze-dry events (CFD) (Fig. 6h–j).

Projected climate-driven changes in winter wheat yield

Across the HHH, future ECEs would consistently lead to reductions in wheat yield (Fig. 7). The comparison of the negative extreme climate effects estimated with the process-based simulated yields and yield statistics reveals that the statistics models do not reproduce the process model-derived negative response of yield to ECEs. For yield changes due to heat stress (EDD) and compound hot-dry extremes (CHD), statistical models generally reproduce the directions of yield change, but the magnitudes of yield change were smaller than estimated from the process-based model, with the largest discrepancies observed in the NS. The ensemble mean of EDD effects suggests that by 2080–2100, warmer climate results in yield declines of 5.6 ± 1.5% (SSP1-2.6), 7.7 ± 2.2% (SSP3-7.0), and 9.4 ± 3.2% (SSP5-8.5) primarily due to temperature stress in spring (EDD_{VP}). However, such yield decline would likely be offset by yield benefits due to reduced exposure to freezing stress in the MYS. Future warming is projected to lead to declines in drought exposure and reduced drought effects. However, future warming would lead to an increase in compound hot-dry events during the reproductive growth phase (CHD_{RP}), primarily driven by substantially rising heat stress, thereby reducing yields. This effect would decrease future wheat yield more in the NS than HHS and MYS sub-regions, and these declining yields were greater as the climate scenarios increased in intensity.

Discussion

Comparison of differences in historical and future extreme climate indices

Temperature plays a multifaceted role in the development and productivity of winter wheat, intricately influencing processes such as vernalization, photosynthesis, and respiration[33,34]. Winter wheat exhibits varying temperature thresholds and sensitivities across its growth phases. Heat stress (EDD), for instance, has been more prevalent in the cooler north sub-region compared to the warmer south sub-region (Fig. 4). Further, drought remains a major constraint across the wheatbelt in China. Climate change-induced water scarcity places huge pressure on agriculture[35–37], with yields highly susceptible to irregular drought stress during critical growth phases[38,39]. Our findings indicate that consecutive dry days (CDD) are projected to decrease overall; however, an increasing trend during the planting to November period was particularly evident in the south sub-region. This aligns with Ji and Kang (2015)[40], who reported that dry events in southern China is predicted to increase by the end of the 21st century. While drought events are projected to remain relatively stable, heat events are expected to rise markedly in the future (Supplementary Fig. 3). Consequently, climate projections suggest that warmer post-anthesis compound hot-dry events (CHD) are primarily driven by rising heat stress, consistent with the findings in Australia, where recurrent drought and increasing heat events contribute to rising frequencies of CHD[41].

Freeze events, defined by diverse temperature thresholds across growth phases, persistently pose substantial challenges for winter wheat. These events were mainly divided into early freeze (germination and dormancy phases) and late freeze (vegetative phase). Between 2000 and 2019, freeze occurrences during dormancy (CFD_{DP}) and vegetative (CFD_{VP}) were frequent. Freeze damage during germination primarily affects the freezing of new leaves and has little impact on the wheat's reproductive organs and yield[42]. However, global warming has led to pronounced warm and cold fluctuations during the stem elongation stage, exacerbating spring freeze events[27]. This was consistent with the previous study, which showed that spring freezes are common in the Australian wheatbelt and regularly cause serious damage to wheat production[43]. However, future projections indicate more reductions in freeze events across the north and much of the wheatbelt sub-regions. Similarly, interactions between drought and freeze events are expected to decline across multiple growth phases, further altering the risk landscape for winter wheat production in Northern China.

Wheat yield sensitivity to extreme climate indices

Large disparities in the wheat yield sensitivity of ECEs were observed across the sub-regions. Projected warming trends in heat stress suggest a greater decrease in yield in the north and middle sub-regions, which is consistent with results found in Australia[41]. Freeze events were not equally distributed during the growing season, and the timing of the occurrence of freeze events should not be neglected when assessing the impact of freeze on crops. The susceptibility of crops to freezing was found to increase with the specific growth phase. The negative sensitivity of wheat to freeze stress was stronger

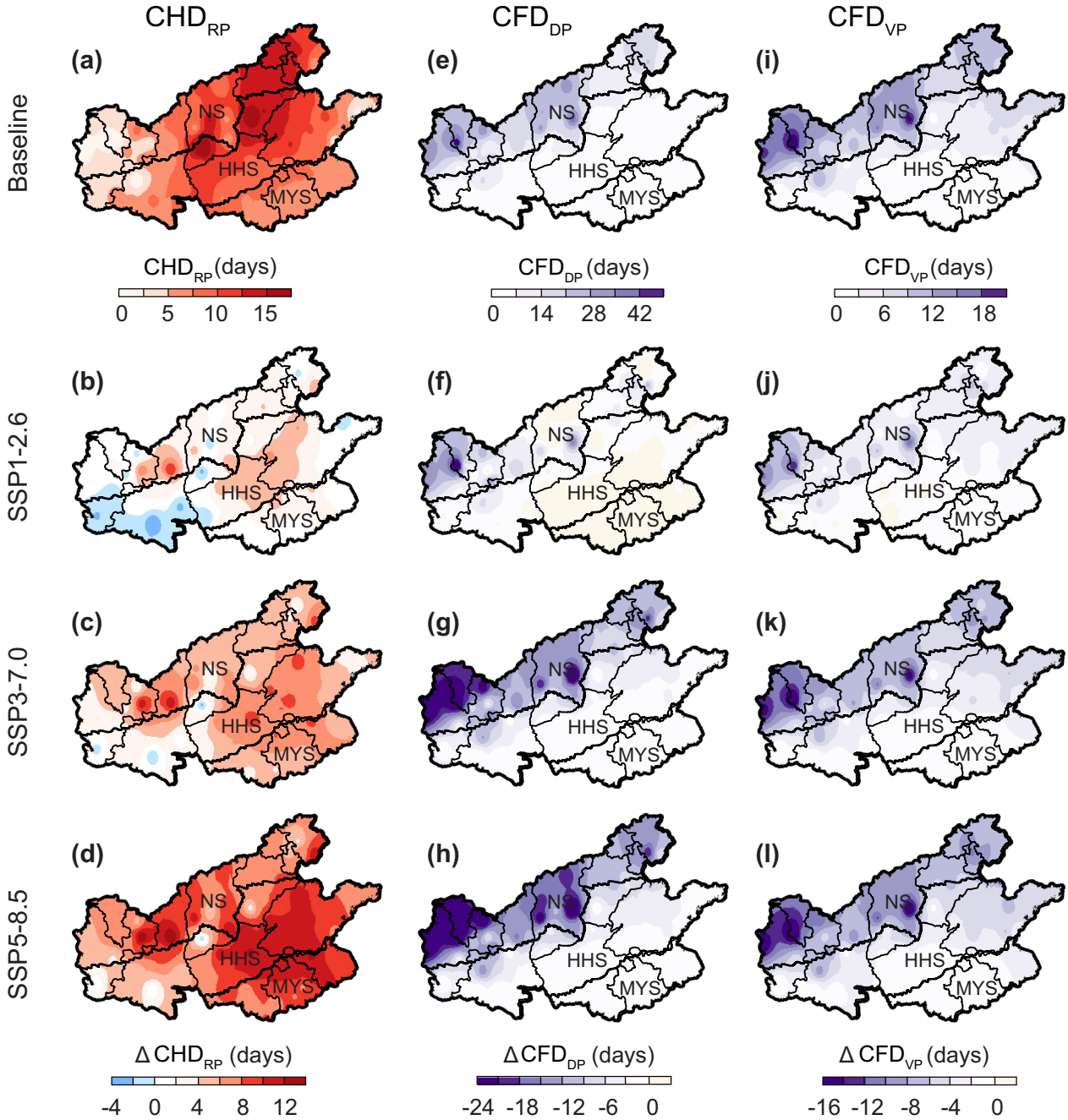


Fig. 5 | Historical and projected changes in compound extreme climate indices. **a–d** Baseline (1999-2019) and projected changes (Δ, 2080–2100) for compound hot-dry days (CHD) during growth phases for each climate scenario. **e–l** Same as (**a–d**) but for compound freeze-dry days (CFD). Future climate projections are based on the ensemble of climate models. DP is the dormancy phase (December to February); VP is the vegetative phase (March to Anthesis); RP is the reproductive phase (Anthesis to Maturity).

in the south sub-region. Thus, warming trends in freeze were beneficial to wheat growth as they reduced cold injury and freeze damage to winter wheat seedlings. Decreasing trends in freeze had positive effects on yield, with spring freeze contributing more to yield reductions. Additionally, we observed that the statistical model tended to predict less negative yield response to temperature extremes than the process-based model. One potential explanation is that statistical models reflect aggregate outcomes across diverse environments, and Northern China is characterized by relatively lower humidity, so that it may experience greater evaporative cooling under warming conditions[44]. Therefore, this would reduce heat impacts. However, this does not necessarily imply the statistical model performed greater accuracy, as the statistical model also captures interactions between temperature and humidity.

Our comparison of yield sensitivities to precipitation extremes revealed additional discrepancies. This divergence is consistent with earlier findings[23], which showed that process-based and statistical models often differ in their estimated precipitation sensitivities. This may be due to the fact that process-based models operate with fixed soil parameters[45] and assumed initial soil water conditions[46], while the statistical models are aggregated across a wide range of field conditions and soil types. Moreover, process-based models may empirically overemphasize the role of precipitation[47] while overlooking other field-level constraints, such as heavy

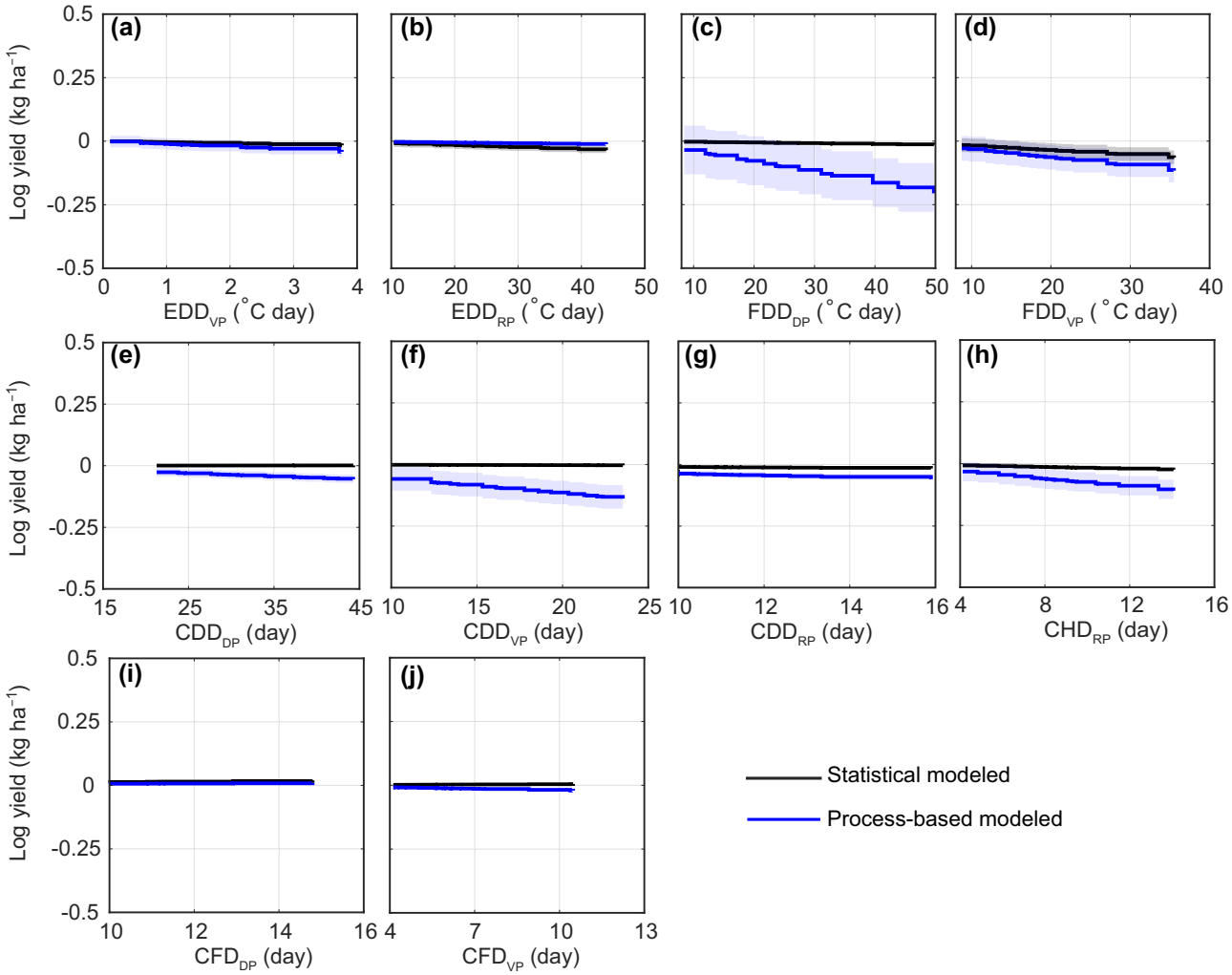


Fig. 6 | Comparison of estimated effects of extreme climate indices on wheat yields. a, b Yield changes based on the process-based crop model and statistical model under extreme degree days (EDD) during the (**a**) VP and (**b**) RP. **c, d** Same as (**a, b**) but for freezing degree days (FDD) during the (**c**) DP and (**d**) VP. **e–g** Same as (**a, b**) but for consecutive dry days (CDD) during the (**e**) DP, (**f**) VP, and (**g**) RP. **h** Same as (**a, b**) but for compound hot-dry days (CHD) during the RP. **i, j** Same as (**a, b**) but for compound freeze-dry days (CFD) during the (**i**) DP and (**j**) VP. DP is the dormancy phase (December to February); VP is the vegetative phase (March to Anthesis); RP is the reproductive phase (Anthesis to Maturity). Black solid lines represent yield responses from the statistical model, and blue solid lines represent those from the process-based model; shaded areas denote 95% confidence intervals.

precipitation within a short time, waterlogging, or lodging. These limitations highlight the value of combining both modeling approaches and point to the need for further study to clarify the underlying causes of divergence.

We also found that climate stresses were often interrelated between water and heat stress. The contributions of temperature anomalies to drought are difficult to disentangle[48,49], and wheat yield could be influenced more by compound ECEs. The negative effects of temperature intertwined with drought were stronger on wheat but exhibited spatial variability within the study region. CHD substantially dominated yield loss in the north subregion. This was because higher spring temperatures might affect wheat yields both through direct heat stress and through associated drought stress[50]. This challenge is expected to intensify with climate change, which is projected to increase the frequency and severity of heat events. In contrast, CFD contributed the most negative effects on wheat yield in the south subregion. Although freeze events are projected to decline overall, drier atmospheric conditions and rapid development of crops may still exacerbate freeze risks, particularly under warmer daytime temperatures and clear nights[51]. These findings highlight the growing challenge of balancing heat and freezing risks to sustain winter wheat production under future climate scenarios.

It should be noted that our analysis explicitly incorporates irrigation, as it might decouple local microclimates and regional climate patterns by altering land–atmosphere feedbacks and atmospheric moisture transport[52,53]. We included irrigation share as a covariate in our statistical models. We find that the estimated impacts of heat (EDD) and dry (CDD) on yield generally remain consistent in the same direction (i.e., negative impact) but are slightly larger in magnitude when irrigation is included, suggesting that omitting irrigation may lead to an underestimation of climate sensitivity. This finding is consistent with Zaveri and Lobell (2019)[54], emphasizing that models excluding irrigation effects tend to provide lower-bound estimates of heat sensitivity.

Uncertainties and limitations

The impacts of ECEs on wheat yield were combined with crops and climate models. Process-based models, including CERES-Wheat, however, have obvious limitations in simulating ECEs. Models usually use the daily mean temperature instead of the daily maximum temperature to evaluate the impacts of heat stress on photosynthesis and grain filling rate; thus, the effect of heat stress on winter wheat yield is generally underestimated[55]. While our analysis focused on the effects of ECEs (heat, freeze, dry, and compound) on

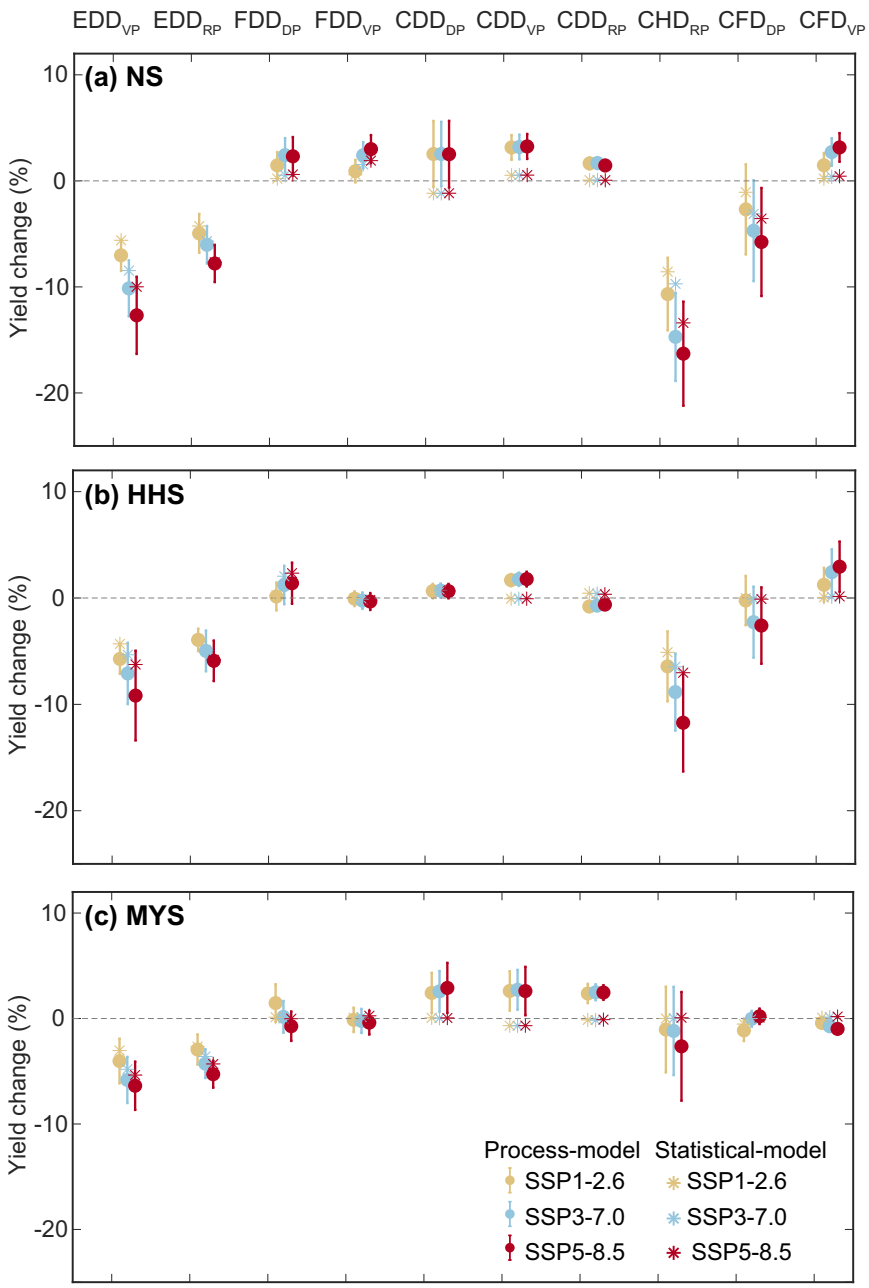


Fig. 7 | Impact of extreme climate indices on wheat yields by 2080-2100 under SSP1-2.6, SSP3-7.0, and SSP5-8.5 scenarios. a Estimated effects of future changes in extreme climate indices on wheat yields, derived from both the process-based and statistical models for the north (NS) sub-region. Extreme climate indices are extreme degree days (EDD), freezing degree days (FDD), consecutive dry days (CDD), compound hot-dry days (CHD), and compound freeze-dry days (CFD). DP is the dormancy phase (December to February); VP is the vegetative phase (March to Anthesis); RP is the reproductive phase (Anthesis to Maturity). **b, c** Same as (**a**) but for (**b**) middle (Huang-Huai, HHS), and (**c**) south (middle-lower reaches of Yangzi River, MYS) sub-regions. Solid circles represent estimated mean coefficients from the process-based model, and asterisks represent those from the statistical model; vertical bars denote the 95% confidence intervals.

wheat yield, however, it is worth noting that crop damage from inundation or flooding was not considered. Given the rising precipitation variability and likelihood of more frequent and intense flooding events under future climate scenarios[56–58], the exclusion of flooding effects may lead to an optimistic estimate of climate-induced yield impacts. In addition, we used consecutive dry days to serve as a drought indicator to connect with heat stress for compound hot-dry events. Low precipitation (consecutive dry days) has been widely adopted in drought assessments, and the use of compound heat-precipitation extreme events are relatively common in China[16,59]. However, the connections between water and heat climate extremes and their biological impacts on crops raise the potential for compound, interactive effects of changing rainfall, temperature, and soil moisture[60,61]. Future studies should incorporate such complementary drought metrics to better capture the complexity of drought impacts on crop productivity.

Conclusion

We quantified the impacts of changes in ECEs on wheat yield in Northern China using the statistical and process-based crop models. The analysis revealed that the magnitude of yield loss due to compound ECEs was greater than that caused by individual climate events. The negative effects of temperature intertwined with drought on wheat yield varied spatiotemporally across three sub-regions. Heat and dry events caused wheat yield loss in the north and middle sub-regions, and yields were further reduced when the compound hot-dry events occurred during the reproductive phase. However, changes in compound freeze-dry events contributed to the most negative effects on wheat in the south sub-region. Future increases in heat stress were projected to cause a yield loss that substantially outweighed the yield gains from decreasing freeze stress. Future projected compound hot-dry extremes led to larger wheat yield reductions in the form of more severe ECEs with longer durations under three climate scenarios. Higher warming

Table 1 | Definitions of extreme degree days (EDD), freezing degree days (FDD), consecutive dry days (CDD), compound hot-dry days (CHD), and compound freeze-dry days (CFD) extreme climate indices

Extreme indices	Index	Definition	Unit
Hot days	HD	The number of days with $T_{max} \geq 30$ °C	d
Extreme degree days	EDD	Accumulated extreme degree days to define heat stress	°C•d
Freeze days	FD	The number of days with $T_{min} \leq T_{frez}$	d
Freezing degree days	FDD	Accumulated freeze degree days to define freezing stress	°C•d
Consecutive dry days	CDD	The maximum number of consecutive days with Precp <1 mm	d
Compound hot-dry days	CHD	Hot days and consecutive dry days simultaneously concurrent	d
Compound freeze-dry days	CFD	Freeze days and consecutive dry days simultaneously concurrent	d

scenarios resulted in more pronounced yield declines in the north sub-region compared to the other sub-regions, making it the primary driver of yield changes in this wheatbelt production area. Our findings suggest that climate warming will exacerbate yield losses, particularly in the north, and highlight the need for targeted adaptation strategies. These results could assist the local winter wheat production community in implementing preventive measures to mitigate the adverse effects of climate change and enhance a better understanding of critical factors to yield improvements.

Methods

Wheat, phenology, and climate data

Winter wheat yield, planted area, and production data at the site level from 1999-2019 were available and retrieved from the National Bureau of Statistics (http://data.cnki.net/). Since most statistical yearbooks do not directly record per-unit yield data, the yield was obtained by dividing the total yield by and planting area. It is worth noting that all yields were calculated in units of metric kilograms per cultivated hectare (kg ha^{-1}).

Winter wheat phenological data, including sowing, anthesis, and maturity, were collected from 1992 to 2013 at 90 agro-meteorological sites operated by the China Meteorological Administration (CMA, http://data.cma.cn/en). To delineate distinct yield sensitivities to climate stresses at different development phases, we partitioned the life cycle of winter wheat into four phenological phases: germination and establishment (planting to November, GEP), dormancy (December to February, DP), vegetative (March to anthesis, VP), and reproductive (anthesis to physiological maturity, RP) phases.

Observed daily climate data, including maximum, minimum, and average temperature, precipitation, and sunshine hours for 1980–2019, were obtained from the China Meteorological Administration (CMA, http://data.cma.cn/en). Daily solar radiation was calculated from observed sunshine hours using the Ångstrom formula[62]. Future climate scenario data of five general circulation models (GCMs) (GFDL-ESM4, MPI-ESM1-2-HR, MRI-ESM2-0, IPSL-CM6A-LR, and UKESM1-0-LL) from 2020 to 2100 were retrieved from the Intersectoral Impact Model Intercomparison Project (ISIMIP, https://data.isimip.org/search/)[63]. We chose ISIMIP because it is bias-adjusted and future climate data from the latest GCMs (CMIP6)[64]. ISIMIP found that climate variables in these five models matched historical observations better and preserved the original warming signal in CMIP6[65]. They can well represent the entire range of GCM ensemble simulations as GFDL-ESM4, MPI-ESM1-2-HR, and MRI-ESM2-0 are low climate sensitivity, while IPSL-CM6A-LR and UKESM1-0-LL are high climate sensitivity[66]. We conducted future projections under 3 Shared Socioeconomic Pathways (SSPs): SSP1–2.6 (less than 2 °C warming; low end forcing pathway), SSP3-7.0 (around 4 °C warming; medium to high-end forcing pathway), and SSP5-8.5 (around 5 °C warming; high end forcing pathway).The climate scenario data were interpolated to the site-level by the inverse distance weighted method with a 0.5° × 0.5° horizontal resolution.

Data for the rainfed and irrigated areas sown with winter wheat for the years 2000 to 2015, with a resolution of 5 arc minutes, were obtained from the global gridded dataset of annual harvested areas for the 21st century (MIRCA-OS)[67]. We finely interpolated grid data to the site-level to compute irrigation share at each location. The average irrigation shares for 2000 to 2015 are shown in Supplementary Fig. 5. Here, the total harvested area of winter wheat was calculated as the sum of the irrigated and rainfed wheat areas. The irrigation share (I) was then the irrigated area divided by the total harvested area at each site. Due to the absence of irrigation data, we assumed no change in irrigation share since 2015.

Extreme climate indices

Five extreme climate indices were used to evaluate the impacts of ECEs on wheat yields (Table 1). Here, the individual ECEs: heat (hot days, HD) captures cumulative extreme degree days (EDD), and is defined as the number of days with T_{max} above 30 °C[68]. For winter wheat in China, the temperature threshold of 30 °C was appropriately used for heat stress, since heat stress over 30 °C may result in reduced stomatal conductance and photosynthetic assimilation, which in turn led to a decline in grain yield formation in Chinese winter wheat[69]. Freeze (freeze days, FD) is the number of days when T_{min} is below T_{frez} during each of the four growth phases, indicating freeze stress occurrence (FDD)[70,71]. The T_{frez} selected represents the freeze injury thresholds associated with each growth phase of wheat in China based on previous studies[33,72], as detailed in Supplementary Table 2. Extreme precipitation deficit is defined as consecutive dry days (CDD) with a daily precipitation <1 mm to characterize drought stress[19]. Bivariate compound extreme events used are compound hot-dry days (CHD) and compound freeze-dry days (CFD), which represent heat stress or freeze stress that occurs concurrently with drought conditions during each growth phase.

Estimating effects of extreme climate indices on wheat yield

Winter wheat growth is subject to the influence of multiple extreme climate indices over distinct seasons. We developed a linear mixed-effects model to quantify the influence of extreme climate indices on wheat yield. These indices include freeze damage caused by extremely low temperatures, as well as drought and heat stresses, all of which possibly contribute to yield reductions[73]. Specifically, we incorporated the aforementioned climate indices, separated based on each growth phase with yield and focused on investigating how distinct extreme climatic stresses affected wheat yields. The model is specified as:

$$\log Y_{d,t} = \sum_{j=2}^{3}\beta_{1,j} FDD^{j}_{d,t} + \sum_{j=3}^{4}\beta_{2,j} EDD^{j}_{d,t} + \sum_{j=2}^{4}\beta_{3,j} CDD^{j}_{d,t} + \beta_{4} CHD_{d,t} + \sum_{j=2}^{3}\beta_{5,j} CFD^{j}_{d,t} + \beta_{6} I_{d,t} + \alpha_{d} + \beta_{7,d} t + \beta_{8,d} t^{2} + \varepsilon_{d,t} \quad (1)$$

where $Y_{d,t}$ is winter wheat yield at site d in year t; the index j denotes one of the different phases of wheat growth season (1 = GEP, germination and establishment; 2 = DP, dormancy; 3 = VP, vegetative; 4 = RP, reproductive); α_d is the site effect to control time-invariant spatial heterogeneity; the transformed yields by logarithms are primarily to stabilize the variance and interpret coefficients in percentage terms as well as to improve the normality and homogeneity of residuals; terms t and t^2 characterize the county-specific nonlinear but progressive technology improvements over time, which are not explained by climate; $\varepsilon_{d,t}$ is the random error; β_{1-5} represents yield response to different extreme climate indices. Since wheat statistical yield data in our study region does not provide information on irrigated and rainfed yields separately, we are unable to directly estimate the impact of weather on each type. Therefore, to measure the change in sensitivity of yields to weather, we include the share of wheat-specific irrigated area ($I_{d,t}$), which we use as our measure of wheat irrigation[54]. All variance inflation factor (VIF) values for climate predictors are summarized in Supplementary Table 3 to confirm that multicollinearity is not a concern in our model.

Crop model input and calibration data

The wheat simulation model used in this study was the CERES-Wheat model, part of the Decision Support for Agrotechnology Transfer (DSSAT). This wheat model has been widely tested and used to study diverse cropping systems around the world, along with variable climates and environments[74,75]. DSSAT is a point-based biophysical modeling framework to simulate crop growth as a function of weather, soil, management, and nutrient availability on a daily step basis. We simulated the wheat growth and yield on a site scale over the HHH wheatbelt for historical and future periods. DSSAT requires the following input data: (I) daily values of maximum and minimum temperature, precipitation, and solar radiation, (II) detailed soil profile data such as soil horizon depths, percentage sand, silt, and clay, bulk density, organic carbon content, pH, and soil water content; (III) management data such as planting date, planting density, planting depth, row spacing, fertilizer practices, irrigation rates, and detailed crop growth parameters. The datasets used for physical and chemical properties of soil data, including soil texture (sand, silt, and clay content), organic carbon, total N, soil bulk density, pH, drained upper limit, lower limit of available water, and saturated volumetric water content in different soil layers, were collected from the China Soil Scientific Database.

We used the CERES-Wheat module to simulate wheat production. Here, irrigation was considered during model simulation. Given the spatially heterogeneous nature of irrigation across the region, the automatic irrigation setting within the model was used to approximate actual field practices. That is, wheat crop yields were simulated based on actual irrigation management practices. Crop parameters are used to set genotype aspects of wheat development. Crop parameters and physiological characteristics are given in the form of seven genetic coefficients (Supplementary Table 4). Four phenology coefficients (P1V, P1D, P5, and PHINT) impact plant phenological development, while the other three growth coefficients (G1, G2, and G3) are related to grain yield. For each winter wheat variety, model calibration was conducted based on the experimental data from one site-year experiment, and model validation was performed with data from other site-years (Supplementary Table 5). A detailed description of the calibration approach is included below. Specifically, the adjustment of the seven parameters (P1V, P1D, P5, G1, G2, G3, and PHINT) combined trial-and-error with DSSAT's Generalized Likelihood Uncertainty Estimation (GLUE) package. First, the parameters were sequentially calibrated through stepwise calibration processes starting with phenological development to obtain a close match between simulated and observed phenology. Then the model was run with these derived crop phenological parameters, and the performance was examined in terms of wheat yields. After calibration, model accuracy was assessed by three statistical indices: mean absolute error (MAE), normalized root mean square error (NRMSE), and Willmott's D-value index of agreement[76], comparing the observed data on anthesis, maturity, and yield during the other years. CERES-Wheat with adjusted genetic parameters exhibited reliable performance for simulating winter wheat phenology from planting to anthesis (ADAT) and planting to maturity (MDAT) (Supplementary Fig. 6). The MAE values were about 4.0 and 4.7 days between simulation and observation for anthesis and maturity days, respectively. Simulated yields showed good agreement with the observations, with an MAE value of 466.7 kg ha^{-1}. The relationship between the observed and simulated yields and phenology dates of winter wheat confirmed that the calibrated CERES-wheat model was robust in reproducing the different yield levels across a wide range of climates in China.

Impacts of future climate change on wheat yield

We used five climate models (GCMs) under three projected scenarios (SSP1-2.6, SSP3-7.0, and SSP5-8.5) to investigate how ECEs will change in the future and their potential impacts on wheat yield. To assess the extent of impacts of future climate change on wheat yield, we considered the restricted scenarios[77]. Specifically, we fixed the agronomic management practices at the baseline level, which represent no adaptation scenarios, while climate data were considered as actual future weather data to be employed in Eq. (1) for three scenarios from 2080 to 2100.

We compared the results of the statistical model with the process-based crop model, which simulated wheat yield for the historical period at a site scale. Specifically, the crop model was driven by the CMA-observed daily climate dataset. Using the climate predictors derived from the historical period, the regression model was fitted to the yield values simulated by the crop model. The resulting series of estimated regression coefficients was then compared to the coefficients originally estimated using observed wheat yield statistics. We further applied the coefficients estimated from the crop and statistics model to GCM climate variables to project yield change. The yield changes projected with the crop model were compared with those projected with the regression model built on wheat yield data. This comparison was used to assess how well crop models simulated the extreme climate indices on wheat yield.

Data availability

In this study, we utilize observed daily climate data, including maximum, minimum, and average temperature, precipitation, and sunshine hours from 1980 to 2019 acquired from the China Meteorological Administration (http://data.cma.cn/en). Future climate scenario data of five general circulation models (GCMs) (GFDL-ESM4, MPI-ESM1-2-HR, MRI-ESM2-0, IPSL-CM6A-LR, and UKESM1-0-LL) from 2080 to 2100 under SSP1-2.6, SSP3-7.0, and SSP5-8.5 were downloaded from the Intersectoral Impact Model Intercomparison Project (ISIMIP) (https://data.isimip.org/search/). The historical agricultural yield data used for our analysis was acquired from the National Bureau of Statistics (http://data.cnki.net/). Winter wheat phenological data were collected from the China Meteorological Administration (http://data.cma.cn/en). Data for the rainfed and irrigated areas sown with winter wheat with a resolution of 5 arc minutes were obtained from the global gridded dataset of annual harvested areas for the 21st century (MIRCA-OS) (https://www.nature.com/articles/s41597-024-04313-w). The observed climate, agricultural yield, and winter wheat phenological data used in this study can be found at: https://zenodo.org/records/14861192.

Code availability

The code used to perform analyses can be found at: https://zenodo.org/records/14861192.

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Author contributions

R.Z. and X.L. designed and coordinated the research. R.Z. processed the data, conducted the analysis, and wrote the initial manuscript draft. R.Z., X.L., G.F.S., S.M.W., and F.Y. interpreted the results and advised on presentation of the main findings. R.Z., X.L., and G.F.S. wrote the manuscript with edits from all other authors.

Competing interests

The authors declare no competing interests.

Additional information

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